

RESEARCH ARTICLE

Surfactants expressed by soil pseudomonads alter local soil–water distribution, suggesting a hydrological role for these compounds

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Abstract

Several biological roles have been demonstrated for surfactants expressed by soil and rhizosphere *Pseudomonas* spp., but the impact of these powerful surface-active agents on the local soil–water distribution within the partially saturated soil pore network has not been examined. To investigate this potential hydrological role, the liquid surface tension (γ)-reducing activities (LSTRA) of 72 pseudomonads isolated from a sandy loam soil by tensiometry of culture supernatants were characterized. Of these, 67% exhibited LSTRA, reducing γ to a minimum (γ_{Min}) of 24 mN m^{-1} established by individual distribution identification analysis. Soil microcosms were then used to examine the impact of surfactant expression on the local soil–water distribution. The volumetric water content (θ) of soil microcosms was significantly lowered ($0.78 \times$) by *Pseudomonas fluorescens* SBW25 expressing the surfactant viscosin compared with a surfactant-deficient mutant ($P < 0.002$). Six of 15 soil pseudomonad isolates examined were found to have a similar impact on θ when compared with sterile microcosms ($P < 0.05$). These findings indicate that surfactant-expressing pseudomonads could modify the local soil–water distributions and that surfactants may therefore play a significant hydrological role in soils, in addition to their recognized biological activities.

Introduction

Bacteria express a range of chemically diverse surface-active agents or surfactants (biosurfactants) (reviewed by Rosenberg, 2006), involved in swarming motility, biofilm formation, antagonistic interactions including broad-range antibacterial activity, inhibition of protist and nematode predation, lysis of fungi and oomycetes and the induction of systemic resistance in plants. The expression of these multi-role surfactants is important for colonization, pathogenicity and biocontrol of plant pathogens, especially among *Pseudomonas* spp., which colonize a wide range of habitats including soil and the rhizosphere (for recent reviews, see D'aes *et al.*, 2010; Raaijmakers *et al.*, 2010).

The cyclic lipopeptides are one class of surfactants expressed by pseudomonads that were initially identified by their antibiotic or biocontrol properties. Cyclic lipopeptides are short oligopeptides with a linked fatty acid tail and their amphiphilic nature and low density mean that they diffuse to the air–liquid (A–L) interface to form a monolayer with

reduced liquid surface tension (γ). *Pseudomonas fluorescens* SBW25 isolated from the sugar beet phyllosphere (Rainey & Bailey, 1996) expresses the nine amino acid, 3-hydroxydecanoic acid-tailed cyclic lipopeptide viscosin (De Bruijn *et al.*, 2007). *Pseudomonas fluorescens* SS101 isolated from the wheat rhizosphere (De Souza *et al.*, 2003) expresses the structurally similar massetolide A (De Bruijn *et al.*, 2008). Both surfactants reduce the γ of experimental cultures to $\sim 30 \text{ mN m}^{-1}$ (De Souza *et al.*, 2003; De Bruijn *et al.*, 2007) and are required for swarming and biofilm development (De Bruijn *et al.*, 2007, 2008). However, viscosin expression may inhibit SBW25 attachment to some surfaces (Moshynets *et al.*, 2011). Massetolide A induces systemic resistance in tomato plants and contributes to root colonization (Tran *et al.*, 2007), while viscosin is known to lyse *Pythium ultimum* and *Phytophthora infestans* zoospores responsible for damping-off and late blight plant diseases, respectively (De Souza *et al.*, 2003; De Bruijn *et al.*, 2007).

Among pseudomonads, the proportion of isolates showing significant liquid surface tension-reducing activity

(LSTRA) appears to be highly variable, with 1–5% of wheat, 18.5% of tomato rhizosphere isolates (De Souza *et al.*, 2003; Hultberg *et al.*, 2008) and 6–60% of soil (Nielsen *et al.*, 2002) isolates found to express surfactants. The lowest γ produced by pseudomonads appears to be in the region of 24–31 mN m⁻¹ (as reported by Hutchison *et al.*, 1995; Nielsen *et al.*, 2002; Bodour *et al.*, 2003; De Souza *et al.*, 2003; Kuiper *et al.*, 2004; De Bruijn *et al.*, 2007; Perneel *et al.*, 2007). Similarly, the archetypal surfactant, surfactin, expressed by the soil bacterium *Bacillus subtilis*, reduces γ to 27 mN m⁻¹ (Peypoux *et al.*, 1999). However, a limit to the surface activity of bacterial surfactants (i.e. γ_{Min}) has not been demonstrated. This may vary between surfactant classes, reflecting different chemical structures, modes of synthesis, expression patterns and biological roles, while identifying γ_{Min} for different groups of bacteria will inform bioprospecting for new surfactants and bioremediation isolates. Determining the statistical distribution of LSTRA and γ_{Min} for different communities is also important for the understanding and modelling of soil–water (moisture) distribution and flow in soils (i.e. through the capillary rise and flow equations), particularly at the micro-to-core (μm –cm) scales, where soil–water γ (γ_{SW}) may vary significantly from site to site depending on resident bacteria.

The impact of reduced γ on liquid surface geometry is significant and can be illustrated by comparing the hemispherical drop of pure water with the irregular flattened shapes produced by liquids with lower γ on a glass slide. Despite this rather dramatic demonstration, little is understood of the local, micro-scale hydrological impact that bacterial surfactant expression may have in soil pore networks. In contrast, our understanding of surfactant activity at larger scales in soil bioremediation and vadose zone hydrology is more advanced (reviewed by Mulligan *et al.*, 2001; Henry & Smith, 2003). Surfactants reduce the capillary pressure, which draws water into the soil pore network. Water distribution within pore networks will also change as less water is retained with lowered γ_{SW} , and differences in γ_{SW} can cause significant flow between adjacent zones (Henry & Smith, 2003). Changes in distribution are reflected by alterations to the water retention curve (also known as the soil–water characteristic) that describes the relationship between soil–water content and applied pressure or matric potential (e.g. Read *et al.*, 2003). Surfactants may also contribute to the uneven wetting, fingering and penetration patterns (preferential flow paths) of soils (reviewed by Morales *et al.*, 2010).

At the micro-scale of the pores inhabited by soil bacteria, a reduced γ_{SW} will increase the concavity and surface area of A–L interfaces when the soil–water is at equilibrium and move the interface back down the pore during drainage. At this scale, water distribution is dominated by capillary forces rather than by gravimetric forces, with water found in

partially saturated pore spaces as small bodies of liquid, thin films and droplets adhering to the pore walls. Water distribution influences microbial niches at the micro-scale by restricting oxygen (O₂) and nutrient diffusion, bacterial communication and motility (Crawford *et al.*, 2005; Or *et al.*, 2007; Young *et al.*, 2008), and may be a key factor differentiating soil bacterial aggregations into dry or partially saturated colonies, A–L interface and fully submerged biofilms. Water infiltration into the soil pore network as pulse events following rainfall impact on plant physiology, soil microbiology and ecosystems (Schwinning *et al.*, 2004), while preferential flow paths through soil can differentiate communities and generate ‘hot spots’ of microbial activity, which in turn may help to establish and maintain these paths (Bundt *et al.*, 2001; Morales *et al.*, 2010). The localized expression of powerful surface-active agents such as surfactants by resident bacteria in partially saturated pores might have a significant effect on micro-scale soil–water distribution and microbial habitats. We suggest here that surfactants may play a significant hydrological role in soils beyond their established biological functions, influencing microbial colonization and activity at the micro-scale and the water retention curve at the core-scale.

The aim of this work was to investigate the potential of soil pseudomonads (SP) to alter soil–water content through LSTR activity in experimental cultures and in soil microcosms. This was achieved by assessing LSTR activity among SP isolates recovered from Labfield sandy loam soil and then, using the same soil to produce repacked soil microcosms, to determine whether soil–water content changes can be attributed to surfactant expression using defined pseudomonad surfactant mutants and a subset of wild-type SP isolates.

Materials and methods

Bacteria and culturing conditions

Wild-type *P. fluorescens* SBW25 (Rainey & Bailey, 1996) and *P. fluorescens* SS101 (De Souza *et al.*, 2003), the surfactant mutants SBW25 Δ viscA (viscosin deficient) (De Bruijn *et al.*, 2007) and SS101 Δ massA (massetolide A deficient) (De Bruijn *et al.*, 2008), 37 *Pseudomonas* spp. from our – 80 °C collection (listed in Supporting Information, Table S1) and 72 pseudomonads isolated from soil (SP-1–72; described below) were used in this study. Bacteria were cultured at 20 °C using King’s B (KB) (King *et al.*, 1954), Luria broth (LB) and M9 (minimal M9 salts with 50 mM glucose) (Sambrook *et al.*, 1989), *Pseudomonas* agar base with cetrinide, fusidin and cephaloridine (PAB-CFC; Oxoid) and soil wash growth (described below) media. Plates contained 1.5% w/v agar. Glycerol stocks were produced by adding 15% v/v glycerol to overnight KB cultures. Cells from

overnight (18 h) KB shaken cultures were washed and resuspended in sterile deionized water to provide inocula. Culture densities and cell numbers were determined by OD_{600 nm} measurements and colony counts, respectively.

Sample site and soil type

Soil was collected from the Labfield site at the James Hutton Institute (Invergowrie, UK) in January 2009. This soil is a eutric cambisol, described as free-draining dark brown, slightly acidic sandy loam with a composition of 6% organic matter, 59% sand, 34% silt and 7% clay (Hallett & Young, 1999).

SPs

Five-gram subsamples ($n = 3$) of soil was homogenized and resuspended in sterile deionized water by vigorous shaking for 30 min. These were allowed to settle before aliquots of the suspension were diluted and spread onto PAB-CFC-selective plates and incubated aerobically at 20 °C to isolate *Pseudomonas* spp. or *Pseudomonas*-like bacteria. The SP collection (SP-1–72) displayed a range of colony phenotypes and included mucoidal (28%) and yellow-fluorescent (31%) isolates, with many also capable of forming biofilms in static KB microcosms (data not shown), suggesting that this collection is heterogeneous.

Soil microcosms and culture medium

Sterile microcosms were produced using air-dried soil sieved to < 2 mm, which was then double autoclaved and packed to 1.4 g cm⁻³ in plastic rings (37 mm id × 40 mm h, sealed at the base with a nylon mesh) according to Pajor *et al.* (2010). Soil washes were prepared by resuspending 25-g portions of soil in 25 mL sterile deionized water by shaking overnight and recovering the supernatant after centrifugation at 3220 g for 10 min. A pooled aliquot was sterilized by autoclaving to provide soil wash growth medium.

Incubating soil microcosms and measuring water content

Microcosms were inoculated with bacteria and incubated at a defined matric pressure (P_M) before assaying the water content. Microcosms ($n = 5$) were placed onto tension tables and saturated from the bottom with sterile deionized water. P_M was adjusted by altering the suction of the table. These were equilibrated at -0.1 kPa and inoculated by the injection of 20 µL of a suspension of $\sim 10^9$ cells 1 cm below the soil surface. After 1 h, P_M was adjusted to -8 kPa, where it was maintained for 7 days at 20 °C before assay (experiments were restricted to a P_M of -8 kPa in response to limited access to tension tables, other resources and time). The inoculation strategy used here is likely to result in the

bacterial colonization of pores with diameters > 25 µm (after White *et al.*, 1994), and preliminary experiments using SBW25 and SS101 indicated that bacteria were able to grow in these microcosms under the conditions described. The water content of soil microcosms was measured as the volumetric water content (θ) (m³ m⁻³), calculated as $(m_W - m_D)/(\rho_W V_b)$, where m_W and m_D are the wet and the dry soil weights (microcosms were dried overnight at 105 °C), ρ_W is the density of water and V_b is the volume of soil before drying.

Determining the water retention curve

The water retention (or release) curve was determined by fitting van Genuchten's curve (van Genuchten, 1980) to volumetric water content measurements of sterile microcosms ($n = 5$) maintained at 20 °C and made over the P_M range 0–10 kPa, which yielded an r^2 value of 0.94. The effect of different γ_{SW} on the water retention curve was predicted using the capillary rise equation after Read *et al.* (2003), assuming that the matric potential (Ψ_m) [P_M equals Ψ_m (J kg⁻¹) expressed on a volume basis] was proportional to γ in an idealized pore space. This approach was used only to produce curves to provide an indication of the impact of surfactants on water retention, but the fitted parameters used here would not describe the full water retention curve for this soil.

Tensiometry

The liquid surface tension (γ) of samples was measured using a Krüss K100 Mk2 Tensiometer at 20 °C as described previously (Koza *et al.*, 2009). The growth of stationary-phase (18 h) cultures ($n = 5$) was first determined by OD_{600 nm} measurements. Cultures were then centrifuged at 3220 g for 10 min to provide cell-free culture supernatants to determine bacterial LSTRA. Here, we define LSTRA as the reduction of the γ of sterile culture medium by ≥ 5 mN m⁻¹. Controls comprised sterile deionized water and culture media; in this work, γ_{Water} was 73.6 ± 0.1 , γ_{KB} 47.0 ± 0.3 , γ_{LB} 41.4 ± 0.5 , γ_{M9} 59.6 ± 1.3 mN m⁻¹ and soil wash (γ_W) was 77.4 ± 0.8 mN m⁻¹. Where necessary, samples and controls were processed in batches, and ANOVA of batch controls was used to determine whether data could be combined for further analysis.

Statistical analyses

Means and SEs are provided where appropriate. Comparison of means was performed using an ANOVA model with Dunnett's *post hoc* comparison test against a control ($\alpha = 0.05$) and Student's *t*-test conducted in JMP v.7.0 (SAS Institute Inc.). Analysis based on general linear models (GLMs) was conducted in SPSS v.18 (IBM Corp.) to

determine whether changes in γ were growth or strain dependent. Individual distribution identification (IDI) analysis in MINITAB v.15 (Minitab Ltd, UK) based on the Anderson–Darling goodness of fit test was then used to identify a theoretical probability distribution to fit the γ data (from a set of 16 standard distributions available in MINITAB) that was further used to predict the minimum γ (γ_{Min}) as indicated by the threshold parameter of the fitted distributions. Finally, van Genuchten's curve was fitted to the volumetric water content using the nonlinear least square procedure available in R v.2.11 (<http://www.r-project.org>).

Results

Determination of the LSTRA of SPs

The LSTRA of 72 SP isolates recovered from a homogenized Labfield sandy loam soil sample is presented in Fig. 1. Thirty-eight (53%) of the SP isolates showed LSTRA (the LSTR group), reducing the γ_{KB} of 47.0 mN m^{-1} to a minimum of $25.2 \pm 0.1 \text{ mN m}^{-1}$ (by SP-1), with a group mean of $28.6 \pm 0.8 \text{ mN m}^{-1}$. Twenty-two of these isolates, from SP-57 to SP-65 in Fig. 1, were not significantly different from that of SP-1 (Dunnett's, $P > 0.05$), suggesting a minimum γ that can be reached by these SPs. Nine SP isolates, from SP-5 to SP-56 in Fig. 1, which appear to reduce γ to intermediate levels, might be expressing surfactants more slowly than the rest of the LSTR group or they may be producing agents with a lower surface activity.

The remaining 34 (47%) SP isolates formed the non-LSTR group with a mean γ of $52.2 \pm 0.4 \text{ mN m}^{-1}$. Five SP isolates, from SP-69 to SP-60 in Fig. 1, had no significant impact on γ_{KB} , while the remainder (40% of the total SP collection) increased it to a maximum of $54.7 \pm 0.3 \text{ mN m}^{-1}$ (SP-43) (Dunnett's, $P > 0.05$). This increase might be attributable to changes in the composition of the medium and the release of cell compounds by lysis during bacterial growth. The inability of these non-LSTR SP isolates to lower γ_{KB} could not be explained by poor or delayed growth in KB medium, as GLM analysis confirmed that growth had no effect on γ ($P = 0.078$), whereas there was a significant isolate-dependent effect ($P < 0.001$).

The efficacy of the LSTRA SP isolates to reduce γ_{KB} was compared directly with *P. fluorescens* SBW25 and SS101, which are known to express surfactants under similar conditions. SS101 was associated with the LSTRA group, but, surprisingly, SBW25 was associated with the non-LSTR SP isolates and did not significantly lower γ_{KB} in this assay (the surfactant-deficient mutants, SBW25 ΔviscA and SS101 ΔmassA , both associated with the non-LSTR SP isolates). However, by extending the incubation of SBW25 from 18 to 24 h, LSTRA was observed, with γ_{KB} reduced to $32.5 \pm 0.4 \text{ mN m}^{-1}$ (neither mutant reduced γ_{KB} during 24-

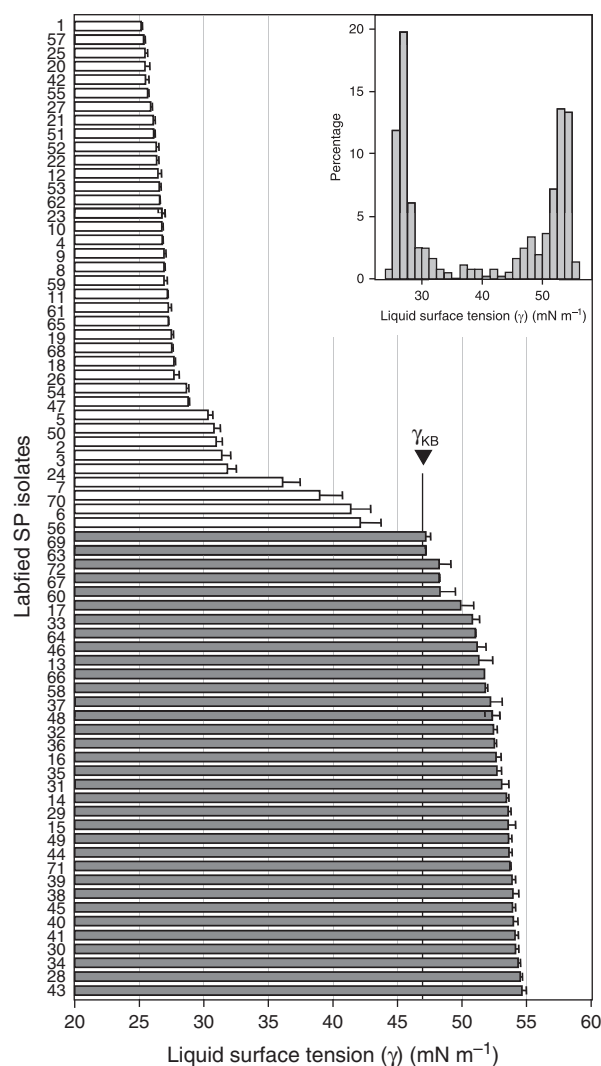


Fig. 1. Many SP isolates demonstrate an ability to reduce liquid surface tension. SP isolates were cultured in KB medium and the liquid surface tension (γ) of cell-free supernatants was determined by tensiometry. Isolates are ranked along the y-axis. Sterile γ_{KB} is indicated by the vertical line; SP isolates showing a clear LSTRA are indicated by white bars (from SP-1 to SP-56); and the rest, shown by grey bars (SP-69 to SP-43), are the non-LSTRA group SP isolates. Mean $\gamma \pm \text{SE}$ are shown. The bimodal distribution of γ is shown by the histogram inset. Data were combined after a comparison of controls indicated that there were no significant differences between batches (ANOVA, $P < 0.05$).

h incubations), demonstrating that small changes in culture conditions can affect the expression of surfactants.

To assess whether non-LSTRA SP isolates might express surfactants under different conditions, 28 of the 35 non-LSTRA SP isolates were tested in LB and M9 media (selected to evenly sample the range of γ produced by the non-LSTRA group) and were compared with SBW25, SBW25 ΔviscA , SS101 and SS101 ΔmassA . Only SP-13 showed LSTRA in LB, reducing γ_{LB} to $25.6 \pm 0.2 \text{ mN m}^{-1}$. However, SP-13 and

seven others showed LSTRA in M9, with the lowest value recorded for SP-13 of $26.3 \pm 0.4 \text{ mN m}^{-1}$. In both LB and M9, SBW25 and SS101 showed LSTRA while SBW25 Δ viscA and SS101 Δ massA did not (data not shown). Once again, for both the LB and the M9 assays, GLM analyses confirmed that growth had no effect on γ (LB, $P = 0.079$; M9, $P = 0.293$), whereas there was a significant isolate-dependent effect ($P < 0.001$) for both media. These findings indicate that 29% (8/28) of the SP isolates not showing a significant LSTRA in KB do so under other growth conditions.

From the KB, LB and M9 tensiometric assays, 67% of the SP isolates were estimated to be capable of significantly reducing γ under one or more growth conditions (53% in KB, 29% of the remaining 47% in LB and/or M9).

IDI analysis predicts the minimum liquid surface tension achievable by SPs

Although SP-1 was found to produce the lowest KB culture supernatant γ , the LSTRA distribution in Fig. 1 suggests that there may be little difference between SP-1 and the next 22 SP isolates. This was confirmed by Dunnett's *post hoc* comparison test using SP-1 as the control ($P = 0.05$), suggesting a limit (γ_{Min}) to LSTRA of $\sim 25 \text{ mN m}^{-1}$ in this assay (this is also supported by the LB and M9 assays, which found no lower value for γ).

IDI analysis was used to determine γ_{Min} and, of the 16 theoretical distributions available in MINITAB, none of the 2-parameter distributions with no lower γ limit fitted the KB tensiometry data. In contrast, three 3-parameter distributions (gamma, log-normal and log-logistic), in which γ was limited from below by a threshold parameter, fitted the data well, the best being the gamma distribution, as indicated by the low Anderson–Darling test statistic value ($AD = 0.688$, $P > 0.05$). This estimated a γ_{Min} threshold value of 24.16 mN m^{-1} , which was in agreement with the log-normal and log-logistic predictions, adding confidence that the true LSTRA γ_{Min} for this collection of SPs is close to 24 mN m^{-1} .

The LSTRA of SPs is similar to that seen for other *Pseudomonas* spp.

In order to determine whether the observed LSTRA distribution was unique to this set of SPs, a further 37 *Pseudomonas* spp. were assayed, from a mixed collection including soil- and plant-associated strains. The distribution of these was similar to that found for the SP collection, with 22 (60%) showing a significant LSTRA and a minimum γ_{KB} of $25.4 \pm 0.1 \text{ mN m}^{-1}$ (the LSTRA of key strains are listed in Table 1). The similarity of the two LSTRA distributions suggests that the ability of the SP isolates to alter the γ of experimental cultures is not unique to this collection, but is a general feature of environmental pseudomonads.

Some pseudomonads can alter the water content of soil microcosms

The impact of SBW25 and SS101 in soil microcosms, in direct comparison with their surfactant-deficient mutants, was determined by investigating whether pseudomonads capable of LSTRA could alter the local soil–water distribution in soil, measured by volumetric moisture content (θ) changes due to the expression of surfactants. Based on the *t*-test results, the mean volumetric water content of SBW25 microcosms was significantly lower ($0.78 \times$) than SBW25 Δ viscA microcosms (SBW25: $0.225 \pm 0.005 \text{ m}^3 \text{ m}^{-3}$; SBW25 Δ viscA: $0.290 \pm 0.007 \text{ m}^3 \text{ m}^{-3}$; $P < 0.002$), indicating that the expression of viscosin resulted in the reduction of water content in these microcosms. In contrast, no significant difference in the volumetric water content was observed between SS101 and SS101 Δ massA microcosms ($P = 0.084$). The failure of SS101 to alter the soil–water content suggests that SBW25 and SS101 may respond differently to the conditions found in the soil microcosms. This is unlikely to be due to nutrient levels, as both SBW25 and SS101, but not the surfactant-deficient mutants, could reduce the γ of soil wash medium (data not shown).

Volumetric moisture content analysis was then extended to a selection of SP isolates and, in these assays, the water content of inoculated microcosms was compared with sterile controls, as surfactant-deficient mutants were not available. Ten LSTR and five non-LSTR group SP isolates were investigated, both sets chosen randomly to cover the range of γ shown in Fig. 1, and the results are shown in Fig. 2. Six of the 15 SP isolates significantly decreased the mean volumetric water content of soil microcosms. Notably, SP-38, which failed to show LSTR activity in KB cultures (but showed LSTRA in M9), was able to alter the water content of these soil microcosms. As the remaining SP isolates did not significantly change the water content compared with the sterile controls, the decrease seen for these six SP isolates could reasonably be attributed to the expression of surfactants, rather than to nonspecific growth-associated activity.

Discussion

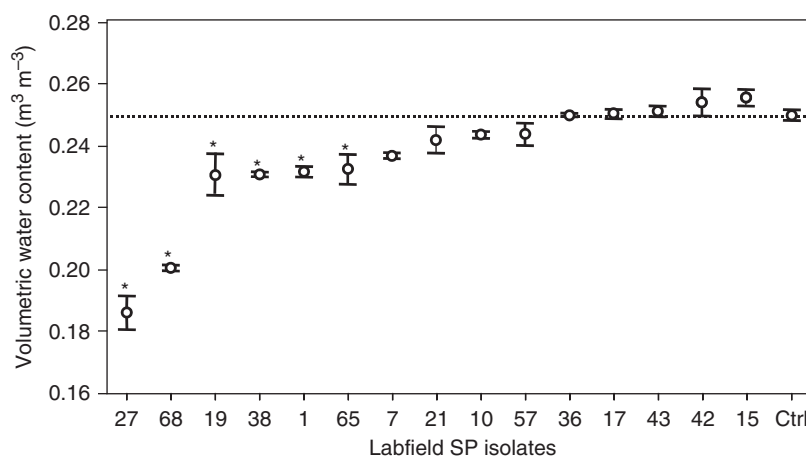
This survey of pseudomonads from Labfield sandy loam soil suggests that at least 67% of isolates are capable of significantly reducing liquid surface tension (LSTR activity) in experimental cultures through the expression of surfactants. This proportion is similar to that found in Højbakkegaard loamy sand (Nielsen *et al.*, 2002), but in general, it appears that the proportion of pseudomonads and total bacteria capable of LSTRA in soils or the rhizosphere is highly variable (recently reviewed by Raaijmakers *et al.*, 2010). It is likely that growth conditions such as nutrient levels and complexity might affect the expression of surfactants, but in this survey, isolates that expressed LSTRA in the nitrogen/

Table 1. Ability of key *Pseudomonas* spp. to reduce the liquid surface tension of KB cultures

<i>Pseudomonas</i> spp.	Activity	γ (mN m ⁻¹)	Origin
<i>P. aeruginosa</i> PA01	LSTRA	32.9 ± 0.2	Opportunistic human pathogen (USA)
<i>P. corrugata</i> NCPPB 2445	LSTRA	39.7 ± 0.2	Tomato pathogen (UK)
<i>P. fluorescens</i> Pf0-1	Non-LSTRA	54.0 ± 0.3	Soil (USA)
<i>P. fluorescens</i> Pf-5	LSTRA	27.9 ± 0.1	Cotton rhizosphere (USA)
<i>P. fluorescens</i> SBW25	LSTRA	32.5 ± 0.4	Sugar beet phyllosphere (UK)
<i>P. fluorescens</i> SS101	LSTRA	25.6 ± 0.2	Wheat rhizosphere (the Netherlands)
<i>P. marginalis</i> NCPPB 2644	Non-LSTRA	51.2 ± 1.6	Alfalfa pathogen (USA)
<i>P. putida</i> KT2440	Non-LSTRA	51.6 ± 0.3	Contaminated soil (Japan)
<i>P. savastanoi</i> NCPPB 64	Non-LSTRA	53.1 ± 0.4	Olive pathogen (Portugal)
<i>P. syringae</i> DC3000	Non-LSTRA	52.5 ± 0.5	Tomato pathogen (Canada)
<i>P. tolaasii</i> NCPPB 2192	LSTRA	29.7 ± 0.2	Mushroom pathogen (UK)

Liquid surface tension (γ) of cell-free KB culture supernatants was determined by tensiometry. All are significantly different from γ_{KB} as determined by a comparison of means using Dunnett's method ($P < 0.0001$). Mean $\pm$ SE shown ($n = 5$ replicates). γ_{KB} is 47.0 ± 0.3 mN m⁻¹. References for the strains not cited in this work are given in Ude *et al.* (2006).

Fig. 2. Volumetric water content of soil microcosms can be altered by pseudomonads. The impact of SP isolates on the local water content was investigated using soil microcosms where the volumetric water content (θ) of inoculated microcosms was compared with sterile controls (Ctrl). Previously, SP-1, 7, 10, 19, 21, 27, 42, 57, 65 and 68 were found in KB cultures while SP-15, 17, 36, 38 and 43 were not. Mean $\theta \pm$ SE are shown. *Significant difference from the Ctrl was tested by a comparison of means using Dunnett's method ($P < 0.05$).



carbon (N/C)-rich KB and LB media also expressed LSTRA in the N/C-simple M9 medium. While the nutrient levels in these are probably 100-fold greater than that found in soil (e.g. Huang *et al.*, 2007), LSTRA was also detectable in SBW25 and SS101 cultures incubated in soil wash medium. This suggests that the range of LSTR activity seen in experimental cultures is also likely to be seen *in situ* among the partially saturated soil pores colonized by these bacteria.

Tensiometry of the LSTR activity of 72 SP isolates allowed the determination of the lower limit for γ reduction (γ_{Min}) of 24 mN m^{-1} by IDI analysis; 25 of the SP isolates produced γ of $\sim 25 \text{ mN m}^{-1}$ and we are not aware of any reports suggesting a substantially lower value achieved by bacterial surfactants. It is possible that the requirements for the synthesis and transport of surfactants out of the bacterial cell, membrane integrity, cell function and survival determine this limit to surface activity.

Analysis of a number of the SP isolates, plus the well-characterized *P. fluorescens* strains SBW25 and SS101, demonstrated that expression of surfactants can alter the water

content in soil microcosms, as determined by the volumetric water content (θ) changes and therefore play a hydrological role in soil and the rhizosphere. These experiments followed an inoculation procedure that would not be expected to result in the colonization of all pore spaces. Nonetheless, significant changes in the water content were observed for SBW25 and six of 15 SP isolates that were tested. Similar microcosms inoculated with SBW25 and SBW25 $\Delta viscA$ have been investigated by microfocus X-ray computed tomography and no structural differences were observed (P.D. Sterpaio & I. Young, unpublished data). This lends weight to our conclusion that the changes in the water content were the result of surfactants acting on water distribution rather than soil structure. Changes in the water content would be expected to vary with colonization patterns and activity. In particular, conditions that allow faster growth and larger populations by providing better nutrient supply and O₂ access might be expected to result in greater levels of surfactant expression and therefore more significant changes in water content. The failure of nine SP isolates to

alter the soil–water content of soil microcosms might be explained by differential growth in soil microcosms, an absence of surfactant genes or the lack of an appropriate environmental signal for surfactant production, or the expression of surfactants with significantly different properties in soil pore networks compared with experimental cultures.

The moisture release curve was calculated for soil microcosms assuming different γ_{SW} values (Fig. 3) to demonstrate how surfactants expressed by SPs might alter the water distribution in soil. The largest impact of reduced γ_{SW} is seen at low P_M (0 to -5 kPa), where less water is predicted to be retained in soil pores; the differences are smaller at higher P_M . Further experimentation is required to test these predictions as, in a similar work, the differences between water retention curves for a plant-derived surfactant in sandy loam microcosms were less than predicted at higher P_M (-4 to -10 kPa), while in acid-washed sand microcosms, the effect was greater than predicted at lower P_M (0 to -7 kPa) (Read *et al.*, 2003). Such differences are likely to be due to the heterogeneous distribution of surfactant, surfactant sorption and hydrophobic pore surfaces, as well as the assumption of an idealized pore space. It is also likely that in real soils, the microbial production of surfactants is likely to affect isolated pores rather than the entire connected network, and the impact of surfactants becomes diluted as soils dry and the surface area of the A–L interfaces increases (Read *et al.*, 2003). The impact of surfactants on γ_{SW} is moderated by sorption to surfaces and depends primarily on the organic content of soil and the chemical nature of the surfactant (Laha *et al.*, 2009).

The observation that some SPs are capable of reducing the volumetric moisture content of soil microcosms through expression of surfactants, and the predicted impact of reduced γ_{SW} on the soil–water distribution, suggest that

bacterial surfactants may play a significant hydrological role in soils beyond their recognized biological functions. The hydrological model (Fig. 4) focuses on the impact that surfactant expression may have on partially saturated pores at equilibrium. Under these conditions, surfactants will act to drain pores and uncover microcolonies or biofilms that may have been O_2 limited. The lowered A–L interface, with increased concavity and surface area, would provide these bacterial aggregations with better access to O_2 and allow further growth. Based on the behaviour of the non-LSTRA SP isolates identified in this work, it is possible that some bacteria may increase γ_{SW} , and raises the possibility that aggregations might antagonistically interact to control the γ_{SW} of local pore spaces. However, we are currently unaware of any other reports of bacteria increasing γ or of mechanisms by which this change might be effected.

Environmental *Pseudomonas* spp., such as SBW25 and many soil and rhizosphere isolates, have a propensity to produce biofilms at the A–L interface of static liquids in which O_2 is accessed from above and nutrients from below (Ude *et al.*, 2006). Access to O_2 in static liquids also drives the emergence of biofilm-forming mutants of SBW25 and explains the fitness advantage that biofilm formers have over non-biofilm-forming strains (Koza *et al.*, 2010). Many of the biofilm-forming *Pseudomonas* spp. from Ude *et al.* (2006) tested here showed LSTR activity (Table 1 and Table S1), while many of the SP isolates are known to produce A–L interface biofilms (A. Koza & A.J. Spiers, unpublished data), suggesting a strong link between biofilm formation and surfactant expression. Surfactants are involved in the initial attachment and biofilm growth of both SBW5 and SS101 (De Bruijn *et al.*, 2007, 2008), as well as in biofilm development and dispersal for a wide range of other bacteria (Raaijmakers *et al.*, 2010). Within soils, biofilm formation, often referred to as bioclogging, can significantly reduce

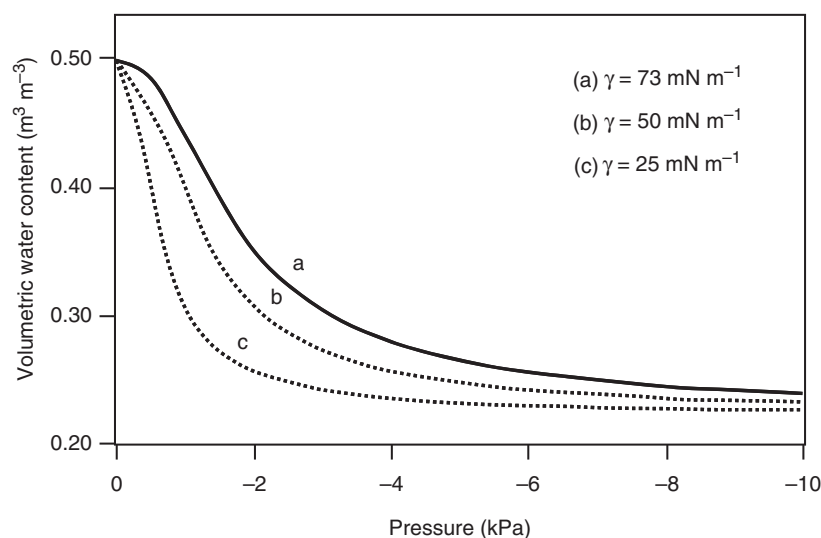


Fig. 3. Reduced soil–water surface tensions alter the soil–water retention curve. The water retention curve determined for soil microcosms assuming γ_{SW} to be ~ 73 mN m $^{-1}$ is shown by (a). The curve is recalculated for γ_{SW} values of 50 mN m $^{-1}$ (b) and 25 mN m $^{-1}$ (c).

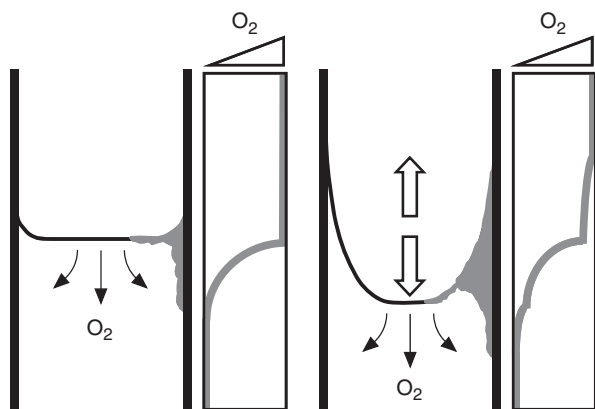


Fig. 4. The reduction of soil–water surface tension (γ_{sw}) in partially saturated soil pores may have a significant ecological impact on colonizing bacteria. Bacteria (grey areas) colonizing the meniscus region and A–L interface of a partially saturated pore at equilibrium are shown in the left schematic. O_2 diffusing into the liquid column is indicated by the small arrows. The respiration of colonists will establish a strong O_2 gradient indicated in the accompanying box. The particular physical–chemical nature of the pore surface and soil–water solution determines the meniscus curvature and the surface area of the A–L interface. Growth of bacteria below the A–L interface is likely to be O_2 limited and reach the stationary phase earlier than those at the interface. The impact of the expression of surfactants by bacteria is shown in the right schematic. Surfactants will alter the shape of the A–L interface, with the top edge moving up and the centre moving down (large arrows) (more active surfactants will have a larger impact than less active or stable surfactants). This will expose more pore surface for colonization, allowing further growth upwards as partially saturated colonies, while the increased concavity and surface area of the A–L interface will allow O_2 to diffuse more deeply into the liquid column, allowing additional growth downwards as biofilms. This new colonization of the pore will result in an extended and more complex O_2 gradient as suggested in the accompanying box.

porosity and hydraulic conductivity as well as promote preferential flow (Morales *et al.*, 2010). Both biofilm matrix and surfactants are microbial legacies that may influence soil structure and hydrology long after the bacteria that produced them have disappeared (Young *et al.*, 2008).

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Authors' contribution

J.F. and A.K. contributed equally to this work.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Table S1. LSTRA of *Pseudomonas* spp. from the mixed collection.

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